

Prime Gaps on the $6N$ Skeleton: An ω -Dependent Shift in the Twin-Prime Gap Distribution, and a Congruence-Lockdown Mechanism

Ruqing Chen

GUT Geoservice Inc., Montreal, Quebec, Canada

ruqing@hotmail.com

May 31, 2026

Abstract

In Part I we showed that the conditional density of twin primes on the $6N \pm 1$ skeleton depends on the factorisation of the centre N through the local enrichment factor $E(N) = \prod_{q|N, q>3} (q-1)/(q-3)$. Here we ask the complementary question: not how many twin centres survive, but how they are *spaced*. Using the exact set of 23,988,173 twin centres in the decade shell S_{10} (where $6N \in [10^9, 10^{10})$) (and the corresponding sets in S_8, S_9), we study the distribution of the gap ΔN between consecutive twin centres, stratified by $\omega_{>3}(N)$, the number of distinct prime factors > 3 of the left centre. The aggregate gap distribution is the known Hardy–Littlewood shape, peaking at the highly divisible values $6\Delta N = 30, 42, 210$. The new finding is that the *shape* shifts systematically with $\omega_{>3}$: the relative preference for the short gap $6\Delta N = 42$ rises monotonically with ω (from 0.73 to 1.55 across $\omega = 1$ –6 in S_{10}) while that for $6\Delta N = 210$ falls (from 1.15 to 0.41); the trend reproduces across S_8, S_9, S_{10} over a $\sim 64\times$ range of sample size. This contradicts the *unconditional* Hardy–Littlewood prediction, in which gap preference depends only on the gap and not on the centre. We identify a first-order mechanism—*congruence lockdown*: each prime factor q of the centre forbids the gap residues $\Delta N \equiv \pm 6^{-1} \pmod{q}$ —which quantitatively reproduces the rise of 42 and 30 and the collapse of the absolute survival of 210, but leaves a genuine residual at high ω . We make no claim about the infinitude of twin primes. The closed-form conditional gap singular series, and the high- ω residual, are posed as open problems.

Keywords: twin primes, prime gaps, Hardy–Littlewood singular series, jumping champions, conditional density, $6N \pm 1$ framework.

1 Introduction

The coarse statistics of prime gaps are classically governed by gap-only laws. The most common gap between consecutive primes up to x —the *jumping champion*—was conjectured by Odlyzko, Rubinstein and Wolf [2] to be 4 and the primorials $2, 6, 30, 210, 2310, \dots$, with 6 reigning until $x \approx 1.74 \times 10^{35}$, where 30 takes over, and 210 only far beyond. Goldston and Ledoan [3] proved, on the Hardy–Littlewood prime k -tuple conjecture, that sufficiently large jumping champions are primorials. The mechanism in all this work is the singular series for a gap d ,

$$\mathfrak{S}(d) = 2C_2 \prod_{\substack{p|d \\ p>2}} \frac{p-1}{p-2}, \quad C_2 = 0.6601\dots,$$

which grows fastest along the primorials and so makes highly divisible gaps the most common. The decisive structural feature is that $\mathfrak{S}(d)$ depends *only on the gap d* : it is blind to where on the number line the gap sits, and in particular to the arithmetic of the integers bracketing it.

In Part I [1] we studied a different, conditional, layer of structure. Restricting to the $6N \pm 1$ skeleton and stratifying centres N by their arithmetic “pressure” $\omega_{>3}(N)$ (the number of distinct prime factors $q > 3$ of N), we derived and validated the local enrichment factor $E(N) = \prod_{q|N, q>3} (q-1)/(q-3)$ governing the conditional probability that $6N \pm 1$ are both prime, and showed it reproduces the squarefree bias of Puskasz [9]. That work answered “how many” twin centres occur at a given factor structure.

The interplay between prime factorisation and gaps has been approached before, along three avenues that we are careful to distinguish from ours. (a) *Empirical twin-gap models*. Kelly and Pilling [4, 5] built, from twin primes below 2×10^{11} , an empirical model of the gap distribution, finding a single unconditional exponential law $\mathcal{P}(s) = m e^{-ms}$ whose decay parameter m depends only on the scale N , and analysing the “high jumper” (most probable separation). Crucially, their separation s is counted *within the sequence of primes* (singletons between twins) and their model carries no dependence on the factorisation of any centre. (b) *Gaps between almost-primes*. Goldston, Graham, Pintz and Yıldırım [6], and subsequently Sono [7], used sieve methods to bound small gaps between E_2 -numbers (integers with exactly two prime factors); there the factor count constrains the *endpoints* of the gap, and the endpoints are almost-primes, not twin primes. (c) *Harmonic/sieve density estimates*. Dolgikh [8] used prime “harmonics” (modulo cycles) to obtain heuristic lower bounds on the absolute density of twin pairs, in service of existence questions. Our study departs from all three: we do not model the unconditional distribution, nor constrain endpoints, nor estimate global density. We perform a conditional, empirical tomography of the gaps between *true* twin pairs, stratifying by the factor count $\omega_{>3}(N)$ of the skeleton *centre* N , and we ask how the gap-distribution *shape* varies across these strata. To our knowledge this conditioning variable—the arithmetic of the centre rather than of the endpoints or the global scale—has not previously been used to resolve the twin-gap distribution.

This paper answers “how far apart”. For consecutive twin centres $N_i < N_{i+1}$ we study the gap $\Delta N = N_{i+1} - N_i$ (in centre-step units; the physical distance between the pairs is $6\Delta N$). The question is whether the gap distribution, like the jumping-champion law, depends only on ΔN , or whether—as Part I might suggest—it also depends on the factor structure of the centres. We find the latter, and we trace it to a concrete conditional mechanism. Three deliberately bounded claims are made:

- (i) the twin-gap distribution exhibits a monotone, scale-stable dependence on $\omega_{>3}(N)$ (§2), which the unconditional Hardy–Littlewood series cannot produce;
- (ii) a first-order *congruence-lockdown* mechanism quantitatively explains most of it—the rise of the short gaps 42, 30 and the collapse of the long gap 210 (§3);
- (iii) a residual at high ω remains, which we pose, with the closed-form conditional singular series, as an open problem (§3).

Nothing here bears on the infinitude of twin primes. As in Part I, the headline phenomenon sits on prior work (the primordial gap preference is the jumping-champion effect of [2]); our contribution is the conditional, ω -resolved layer and its mechanism.

Data. All statistics come from an exhaustive scan of the $6N \pm 1$ skeleton with complete factorisation and deterministic interval-sieve primality; the S_{10} twin count 23,988,173 matches Part I to

the integer. Gaps spanning a shell boundary are discarded so that $\log(6N)$ is effectively constant within each stratum. Code and data are released openly [11].

2 The ω -dependence of the gap distribution

2.1 The aggregate distribution is the known shape

Aggregated over all centres, the gap distribution in S_{10} peaks exactly where the jumping-champion law predicts: the most frequent gaps are $6\Delta N = 30$ (2.45%), 42 (2.28%), then 90, 60, 210, ...—the highly divisible values. We confirmed (Part II companion computation) that this aggregate shape is reproduced to correlation 0.999 by the two-endpoint Hardy–Littlewood correlation factor $C_2(\Delta N)$ times an ω -independent geometric envelope. At the aggregate level there is nothing new: the primordial preference is the known effect.

2.2 The shape shifts with $\omega_{>3}$

The new structure appears on stratifying by $\omega_{>3}(N)$. Define the stratified residual $r(g \mid \omega) = P_{\text{obs}}(g \mid \omega) / P_{\text{agg}}(g)$, the gap frequency in stratum ω relative to the aggregate. Figure 1 shows the row-normalised gap distribution by stratum in S_{10} (only strata with $\geq 10^4$ gaps are shown; $\omega \geq 7$ is omitted as its sample— $\sim 1.2 \times 10^3$ gaps at $\omega = 7$ and negligibly many at $\omega = 8$ —is dominated by Poisson noise and supports no reliable claim).

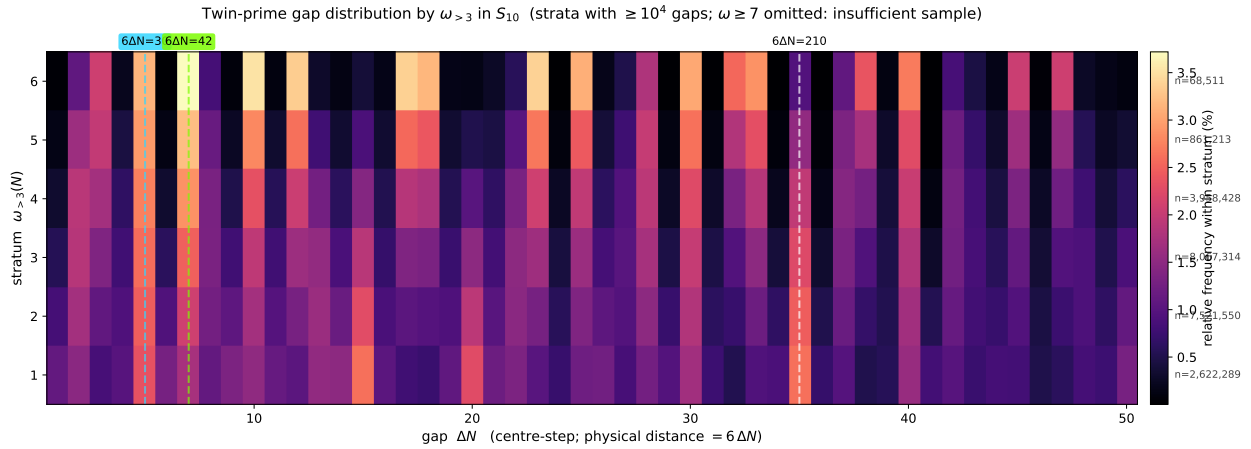


Figure 1: Twin-gap distribution by stratum $\omega_{>3}$ in S_{10} (row-normalised; strata with $\geq 10^4$ gaps). The $6\Delta N = 210$ column (rightmost guide) is bright at low ω and fades upward, while the $6\Delta N = 42$ column brightens upward. Sample sizes per stratum are annotated.

The two principal gaps move in opposite directions, monotonically (Table 1): the preference for $6\Delta N = 42$ rises with ω and that for 210 falls. The effect is not subtle—each spans roughly a factor of two—and it is stable across three independent shells differing by $\sim 64\times$ in sample size.

2.3 Controls

The effect is not an artefact of the three obvious confounders, each checked explicitly:

- (1) **Not noise:** reproduces across S_8, S_9, S_{10} (Table 1) with near-identical values.

$6\Delta N$	shell	$\omega=1$	2	3	4	5	6
42	S_8	0.72	0.90	1.12	1.31	1.54	–
	S_9	0.73	0.90	1.06	1.30	1.42	1.38
	S_{10}	0.73	0.88	1.05	1.22	1.37	1.55
210	S_8	1.14	1.06	0.94	0.83	0.53	–
	S_9	1.14	1.07	0.98	0.82	0.61	0.21
	S_{10}	1.15	1.07	0.99	0.87	0.68	0.41

Table 1: Stratified residual $r(g \mid \omega)$ at the two principal gaps, across three shells. Monotone, opposing, and scale-stable.

- (2) **Not a size effect:** re-normalising each stratum by its own fitted geometric envelope (absorbing that stratum’s mean gap) does not flatten the trend; the residual span is unchanged or slightly larger.
- (3) **Not merely the Part I scalar density:** the fitted gap-decay rate a_ω *decreases* with ω while the twin density rate *increases*, so a_ω is not a function of density alone ($a_\omega/\rho_\omega : 0.32 \rightarrow 0.11$).

Crucially, the unconditional Hardy–Littlewood series cannot produce *any* of this: $\mathfrak{S}(\Delta N)$ depends only on ΔN . The shift must come from a conditional mechanism, to which we now turn.

3 Mechanism: congruence lockdown, and an open problem

The ω -dependence established in §2 contradicts the unconditional Hardy–Littlewood picture, in which the gap singular series $\mathfrak{S}(d) = 2C_2 \prod_{p \mid d, p > 2} \frac{p-1}{p-2}$ depends only on the factorisation of the gap d and is blind to the centre. The resolution is that the relevant object here is not the unconditional series but a *conditional* one: the centre N carries divisibility information that filters which gaps can reach a second twin centre. We make this precise, test it against the data, and isolate exactly the part that remains unexplained.

3.1 The congruence-lockdown factor

Fix a prime $q > 3$. A centre M hosts a twin pair only if $6M \pm 1 \not\equiv 0 \pmod{q}$, i.e. $M \not\equiv \pm 6^{-1} \pmod{q}$; call $\{+6^{-1}, -6^{-1} \pmod{q}\}$ the two *fatal* classes for q . Now suppose the left centre N is a twin centre and consider a candidate gap d , so the right centre is $N + d$.

Proposition 1 (lockdown condition). *If $q \mid N$, then the right centre $N + d$ avoids the fatal classes for q if and only if*

$$d \not\equiv \pm 6^{-1} \pmod{q}.$$

Proof. $q \mid N$ gives $N \equiv 0$, so $N + d \equiv d$. The right centre is fatal for q exactly when $d \equiv \pm 6^{-1} \pmod{q}$; otherwise $6(N + d) \pm 1 \not\equiv 0 \pmod{q}$. (The left centre is automatically safe, since $0 \not\equiv \pm 6^{-1} \pmod{q}$ for $q > 3$.) $\square$

Thus every prime factor q of the centre N imposes a *congruence lockdown* on the admissible gaps: it forbids the two residues $d \equiv \pm 6^{-1} \pmod{q}$. A centre rich in distinct small prime factors (large $\omega_{>3}$) accumulates many such locks simultaneously, and a gap survives to the next twin centre only if it threads all of them. This is the conditional mechanism absent from the unconditional series.

For a stratum of centres with a given $\omega_{>3}$, the lockdown survival of a gap d is the fraction of those centres whose small-prime factor set contains *none* of the primes that lock d :

$$L_\omega(d) = \Pr_{N: \omega_{>3}(N)=\omega} \left[q \nmid N \text{ for every } q > 3 \text{ with } d \equiv \pm 6^{-1} \pmod{q} \right].$$

The predicted gap shape in the stratum is then $\propto C_2(d) L_\omega(d)$, with $C_2(d)$ the (centre-independent) two-endpoint correlation factor of §2.

3.2 The mechanism is confirmed for the short orbits and for the absolute collapse of 210

We evaluate $L_\omega(d)$ directly from the S_9 data (2.98×10^6 twin centres), using the small primes $q \in \{5, 7, \dots, 47\}$ that carry essentially all lockdown weight for $d \leq 60$, and compare $C_2(d)L_\omega(d)$ (normalised within each stratum) to the observed $r(g | \omega)$.

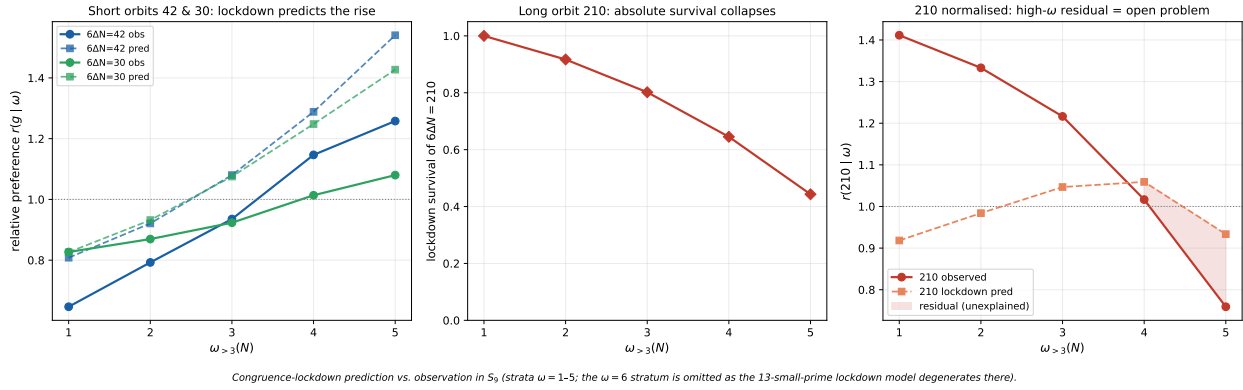


Figure 2: Congruence-lockdown prediction versus observation in S_9 (strata $\omega = 1-5$; the $\omega = 6$ stratum is omitted because the finite 13-small-prime model degenerates there). **Left:** for the short, highly divisible gaps $6\Delta N = 42$ and 30, the lockdown prediction reproduces the monotone rise with ω in both direction and approximate magnitude. **Centre:** the lockdown survival $L_\omega(210)$ of the long gap $6\Delta N = 210$ falls from 1.0 to 0.44 across $\omega = 1-5$: centres rich in factors are increasingly *forbidden* from reaching a twin partner exactly 210 away. **Right:** after normalisation the predicted and observed $r(210 | \omega)$ agree through $\omega \approx 4$ but separate at high ω , leaving a residual (shaded).

Two of the three principal features are quantitatively explained. Before turning to them we record an equivalent, and more transparent, form of the lockdown condition. The fatal congruence $d \equiv \pm 6^{-1} \pmod{q}$ of Proposition 1 is equivalent to

$$q \mid (6d - 1) \quad \text{or} \quad q \mid (6d + 1).$$

Thus a prime factor q of the centre locks the gap d *precisely when* q divides one of the two flanking integers $6d \pm 1$. The admissibility of a gap is therefore governed by the factorisation of $6d \pm 1$ themselves—a clean arithmetic criterion that makes the asymmetry between the principal gaps immediate:

- For $6\Delta N = 210$ ($d = 35$): $6d \mp 1 = 209, 211$, and $209 = 11 \times 19$. The long gap is flanked by a number with two *small* prime factors, so it is locked by the common primes 11 and 19.
- For $6\Delta N = 42$ ($d = 7$): $6d \mp 1 = 41, 43$, both *prime*. The short gap is locked only by 41 and 43.

Since a random centre is divisible by 11 or 19 with probability $\frac{1}{11} + \frac{1}{19} \approx 0.14$, but by 41 or 43 only with probability $\frac{1}{41} + \frac{1}{43} \approx 0.05$ —about three times rarer—factor-rich centres suppress the gap 210 far faster than the gap 42. This furnishes a quantitative, first-principles account of the *asymmetry* in the evolution rates of the two orbits, not merely their directions.

With this in hand the two confirmed features read as follows:

- **Rise of the short orbits (42, 30).** The lockdown prediction rises monotonically with ω , tracking the observed $r(42 \mid \omega) : 0.73 \rightarrow 1.42$ and $r(30 \mid \omega) : 0.91 \rightarrow 1.19$ in both sign and approximate size (Fig. 2, left). Being flanked by primes (41, 43) or large factors, the short gaps are locked by rare divisors and so survive preferentially as ω grows.
- **Collapse of the long orbit (210).** The absolute lockdown survival $L_\omega(210)$ falls by more than half across the strata (Fig. 2, centre). Because $6 \cdot 35 - 1 = 209 = 11 \times 19$ is built from small primes, factor-rich centres very frequently divide it and are thus barred from this gap. This is the origin of the observed decline of 210.

We note that this vindicates the lockdown picture against a natural objection: one might expect the forbidden residues $d \equiv \pm 6^{-1} \pmod{q}$ to fall on gaps indifferently and so produce no systematic preference. They do not: the short, highly divisible gaps are systematically locked by fewer small primes than long gaps of comparable singular-series weight, so the filtering is genuinely selective. This selectivity is confirmed by the data, not assumed.

3.3 The residual, and the open problem

The agreement is not complete. After normalisation, the observed $r(210 \mid \omega)$ continues to fall at high ω (to 0.61 at $\omega = 5$) while the first-order lockdown prediction levels off near 1.0 (Fig. 2, right). A real residual survives at high arithmetic pressure.

We do not paper over this. The first-order lockdown factor $L_\omega(d)$ treats the locks from distinct primes as independent and ignores two genuine higher-order effects: (i) correlations between the factorisations of N and $N + d$ beyond the single divisibility $q \mid N$ used in Proposition 1; and (ii) the Part I density enrichment $E(N)$ of the *left* centre, which is itself ω -dependent and modulates the overall normalisation. A complete account requires the conditional gap singular series

$$\mathfrak{S}_\omega(d) = (\text{joint two-centre singular series, conditioned on } \omega_{>3}(N) = \omega),$$

that is, the Hardy–Littlewood density for the quadruple $\{6N - 1, 6N + 1, 6N + 2d - 1, 6N + 2d + 1\}$ averaged over centres of fixed ω , in the spirit of the singular-series averages of Odlyzko–Rubinstein–Wolf and Goldston–Ledoan for consecutive-prime gaps. We have not obtained this in closed form, and several plausible first-order derivations we attempted do not reproduce the high- ω residual.

Open problem. Derive the conditional twin-gap singular series $\mathfrak{S}_\omega(d)$ and show whether it accounts for the high- ω residual in $r(210 \mid \omega)$, or whether a genuinely new higher-order correlation between neighbouring twin centres is required.

What the data establish, and what we claim, is therefore bounded and precise: the ω -dependence of the twin-gap distribution is real, reproducible across S_8 – S_{10} , contradicts the unconditional Hardy–Littlewood series, and is *predominantly*—but not entirely—explained by the first-order congruence lockdown of Proposition 1. The unexplained high- ω residual, together with the closed-form conditional singular series, is left as a well-posed problem for analytic number theory.

4 Conclusion

Part I established that the factorisation of the centre controls *how many* twin primes occur, through the scalar enrichment $E(N)$. This paper shows it also controls *how they are spaced*: the twin-gap distribution carries a monotone, scale-stable dependence on $\omega_{>3}(N)$ that the unconditional Hardy–Littlewood/jumping-champion framework—gap-only by construction—cannot generate. We traced the effect to a first-order congruence-lockdown mechanism, in which each prime factor of the centre forbids two gap residues, and verified against S_9 data that it reproduces the rise of the short primorial gaps 42 and 30 and the collapse of the absolute survival of 210. A real residual at high arithmetic pressure remains beyond the first-order mechanism; together with the closed-form conditional gap singular series $\mathfrak{S}_\omega(\Delta N)$, it is left as a well-posed problem. The arc from Part I to Part II is thus from a local density law to its imprint on the spacing structure—a conditional, factor-resolved refinement of the classical gap statistics, demonstrated on $\sim 2.4 \times 10^7$ twin centres and offered, with its open residual, to analytic number theory.

Data and code availability. All twin-centre data, verification scripts (gap statistics, Hardy–Littlewood comparison, size and density controls, and the congruence-lockdown model), and the figure-generating code are openly available [11]; the full scan is reproducible from a single segmented-sieve program with deterministic primality, self-verified against an independent factoriser.

References

- [1] R. Chen, *Prime distribution on the $6N$ skeleton: a factor-resolved local model for the conditional density of twin primes*, (Part I), Zenodo, doi:10.5281/zenodo.20470367 (2026).
- [2] A. Odlyzko, M. Rubinstein, M. Wolf, *Jumping champions*, Experiment. Math. **8** (1999), 107–118.
- [3] D. A. Goldston, A. H. Ledoan, *Jumping champions and gaps between consecutive primes*, Int. J. Number Theory **7**(6) (2011), 1413–1421; doi:10.1142/S179304211100471X; arXiv:0910.2960.
- [4] P. F. Kelly, T. Pilling, *Characterization of the distribution of twin primes*, arXiv:math/0103191 (2001).
- [5] P. F. Kelly, T. Pilling, *Implications of a new characterisation of the distribution of twin primes*, arXiv:math/0104205 (2001).
- [6] D. A. Goldston, S. W. Graham, J. Pintz, C. Y. Yıldırım, *Small gaps between primes or almost primes*, Trans. Amer. Math. Soc. **361**(10) (2009), 5285–5330; MR2515812; doi:10.1090/S0002-9947-09-04788-6; arXiv:math/0506067.
- [7] K. Sono, *Small gaps between the set of products of at most two primes*, J. Math. Soc. Japan **72**(1) (2020), 81–118; doi:10.2969/jmsj/74937493; arXiv:1605.02920.
- [8] S. Dolgikh, *Prime harmonics and twin prime distribution*, Cambridge Open Engage, working paper, version 2 (2020), doi:10.33774/coe-2020-c9x75-v2.
- [9] W. Puzskar, *Statistical bias in the distribution of prime pairs and isolated primes*, arXiv:1807.00406 (2018).
- [10] G. H. Hardy, J. E. Littlewood, *Partitio Numerorum III*, Acta Math. **44** (1923), 1–70.

- [11] R. Chen, *6N twin-prime gap data and code* (Part II), <https://github.com/Ruqing1963/6N-twin-prime-gap-omega-dependence> (2026).